

SCIENCE EDUCATION

PHYSICS

A Comprehensive Guide for Class XI

Premium Edition • CBSE Curriculum

Units and Measurement

The Foundation of Quantitative Science

Physics is fundamentally a science of measurement. Every physical phenomenon, from the motion of planets to the flow of electric current, must be quantified before it can be understood, analyzed, or predicted. The language of physics is written in numbers and units, making measurement the cornerstone upon which the entire edifice of physical science rests.

In our daily lives, we constantly make measurements — checking the time, weighing ingredients for cooking, or measuring distances while traveling. However, in scientific inquiry, measurement takes on a more rigorous and systematic form. Scientists must ensure that their measurements are not only accurate but also reproducible by others anywhere in the world. This is where the concept of standard units becomes crucial.

This chapter introduces you to the fundamental principles of measurement, including different systems of units, the international system of units (SI), accuracy and precision, significant figures, dimensional analysis, and error estimation. These tools will serve you throughout your study of physics and beyond, forming the basis for all quantitative scientific work.

Historical Background: The Story of Measurement

Measurement is as old as civilization itself. The **ancient Egyptians (~3000 BC)** used the *cubit* — the length of the Pharaoh's forearm from elbow to fingertip (~52 cm) — to build the pyramids with astonishing precision. The **Babylonians** divided the day into 24 hours and the circle into 360 degrees. The **Greeks**, led by **Eratosthenes (~240 BC)**, measured the circumference of the Earth using shadows and geometry — getting within 2% of the modern value!

For centuries, however, measurement was chaotic. Every town, every kingdom had its own units. A "foot" in Paris was different from a "foot" in London. This caused enormous problems in trade, science, and engineering.

The breakthrough came during the **French Revolution (1790s)**. The revolutionary government commissioned the **French Academy of Sciences** — including luminaries like **Lagrange** and **Laplace** — to create a rational, universal system of measurement. They defined the **metre** as one ten-millionth of the distance from the North Pole to the Equator through Paris, measured by a painstaking survey (1792–1799). This became the foundation of the *metric system*.

In **1960**, the *General Conference on Weights and Measures (CGPM)* established the **SI system** with seven base units. Most recently, the **2019 redefinition** anchored all SI units to exact values of fundamental physical constants (speed of light, Planck constant, etc.), making them independent of any physical artefact — a metre is no longer defined by a metal bar in Paris, but by the speed of light itself.

1.1 Basics of Measurement

1.1.1 Physical Quantities

Definition

Physical Quantity: A physical quantity is a characteristic or property of an object or phenomenon that can be measured and expressed as a number together with a unit. For example, the length of a table, the mass of a book, or the temperature of water are all physical quantities.

Every measurement has two essential components: a **numerical value (magnitude)** and a **unit**. Saying "the distance is 5" is meaningless without specifying whether it's 5 meters, 5 kilometers, or 5 inches. The unit provides the scale or standard against which we compare the quantity being measured.

STRUCTURE OF A PHYSICAL QUANTITY

$$\text{Physical Quantity} = \text{Numerical Value (n)} \times \text{Unit (u)}$$

For example, length of a room = 5 m. Here, 5 is the numerical value and m (metre) is the unit. Note that if we change the unit, the numerical value also changes: 5 m = 500 cm = 0.005 km. The product $n \times u$ always remains constant.

Classification of Physical Quantities

- **Scalar quantities:** Have only magnitude (e.g., mass, temperature, speed, distance, time, energy)
- **Vector quantities:** Have both magnitude and direction (e.g., velocity, displacement, force, acceleration, momentum)

1.1.2 Units and the Concept of Measurement

Definitions

Measurement: Measurement is the process of comparing an unknown physical quantity with a known fixed standard quantity of the same kind. The known fixed standard is called a *unit*.

Unit: A unit is a conventionally accepted standard of reference used to express a physical quantity. It must be well-defined, easily reproducible, internationally accepted, and should not change with time, place, or physical conditions.

Understanding Measurement

When we say the length of a rod is 3 metres, it means the length of the rod is 3 times the standard length called "metre." Here:

- The **standard** = 1 metre
- The **numerical value** = 3 (how many times the standard fits into the quantity)
- The **measurement** = 3 m

1.1.3 Need for Measurement

Measurement is essential in science and daily life for the following reasons:

Why Do We Need Measurement?

- **Quantitative description:** Without measurement, we can only describe phenomena qualitatively ("hot," "fast," "heavy"). Measurement gives precise, quantitative descriptions ("98.6°F," "120 km/h," "70 kg")
- **Verification of laws:** Physical laws like $F = ma$ or $E = mc^2$ can only be verified through precise measurements
- **Reproducibility:** Standardized measurements allow experiments to be repeated and verified by scientists across the world
- **Technological progress:** Engineering, medicine, construction, and all technology depend on precise measurements
- **Communication:** Standard units provide a universal language for exchanging scientific information
- **Prediction:** Accurate measurements enable us to predict future events (eclipses, weather, orbits)

The Measurement Process



Examples of Measurement

Mass of apple = 0.2 kg	Length of table = 1.5 m
Time of race = 10.5 s	Electric current = 2.5 A
Temperature = 37°C	Speed of car = 60 km/h

Figure 1.1: The process of measurement — comparing an unknown quantity with a standard unit

1.1.4 Fundamental and Derived Quantities

Physical quantities are classified into two broad categories based on whether they can be defined independently or require other quantities for their definition.

Definitions

Fundamental (Base) Quantities: Physical quantities that cannot be expressed in terms of other physical quantities. They are independent and form the foundation of the measurement system. There are seven fundamental quantities in the SI system.

Derived Quantities: Physical quantities that can be expressed as combinations of fundamental quantities through mathematical relationships (multiplication, division, etc.). For example, velocity = length / time.

Real-Life Example

When you check the weather forecast, you encounter several fundamental quantities: the temperature might be 25°C (temperature), the time for sunrise could be 6:30 AM (time), and if you're planning to jog, you might plan to cover 5 km (length) in 30 minutes (time). Your running speed ($5 \text{ km} / 0.5 \text{ h} = 10 \text{ km/h}$) is a *derived* quantity.

1.2 Systems of Units

Over the centuries, different countries and scientific communities developed their own systems of units. A **system of units** is a complete and consistent set of units for measuring all physical quantities. The system defines base units for fundamental quantities, from which all derived units are obtained.

1.2.1 CGS System (Centimetre-Gram-Second)

Developed in France, the CGS system uses the **centimetre** as the unit of length, the **gram** as the unit of mass, and the **second** as the unit of time. It was widely used in scientific work during the 19th and early 20th centuries. Many older physics textbooks and some branches of science still use CGS units.

CGS Derived Units

- Force: $1 \text{ dyne} = 1 \text{ g cm s}^{-2}$ ($1 \text{ N} = 10^5 \text{ dyne}$)
- Energy: $1 \text{ erg} = 1 \text{ g cm}^2 \text{ s}^{-2}$ ($1 \text{ J} = 10^7 \text{ erg}$)
- Pressure: $1 \text{ barye} = 1 \text{ dyne/cm}^2$



Carl Friedrich Gauss — Architect of the CGS System

1777–1855, Germany

Often called the "Prince of Mathematicians," Gauss proposed the CGS system in 1832 for magnetic measurements. He argued that all physical quantities could be expressed in terms of just three base units: centimetre, gram, and second. This was the first systematic attempt at a coherent system of units and influenced all subsequent developments, including the modern SI system.

1.2.2 FPS System (Foot-Pound-Second)

The FPS system, also called the British Imperial system, uses the **foot** as the unit of length, the **pound** as the unit of mass, and the **second** as the unit of time. It is still commonly used in the United States for everyday measurements but has been largely replaced by SI in scientific work.

1.2.3 MKS System (Metre-Kilogram-Second)

The MKS system uses the **metre** as the unit of length, the **kilogram** as the unit of mass, and the **second** as the unit of time. It formed the basis for the modern SI system and is very close to SI for mechanical quantities.

Property	CGS System	FPS System	MKS System
Length	centimetre (cm)	foot (ft)	metre (m)
Mass	gram (g)	pound (lb)	kilogram (kg)
Time	second (s)	second (s)	second (s)
Force	dyne	poundal	newton (N)

Property	CGS System	FPS System	MKS System
Energy	erg	foot-poundal	joule (J)
Usage	Older scientific work	USA, UK (everyday)	Basis of modern SI



Giovanni Giorgi — The Bridge from MKS to SI

1871–1950, Italy

In 1901, Italian physicist Giovanni Giorgi proposed extending the MKS system by adding the ampere as a fourth base unit for electrical quantities. His MKS-Ampere (MKSA) system elegantly unified mechanical and electrical units, eliminating awkward conversion factors. This proposal, initially controversial, was eventually adopted and became the foundation of the modern SI system established in 1960.

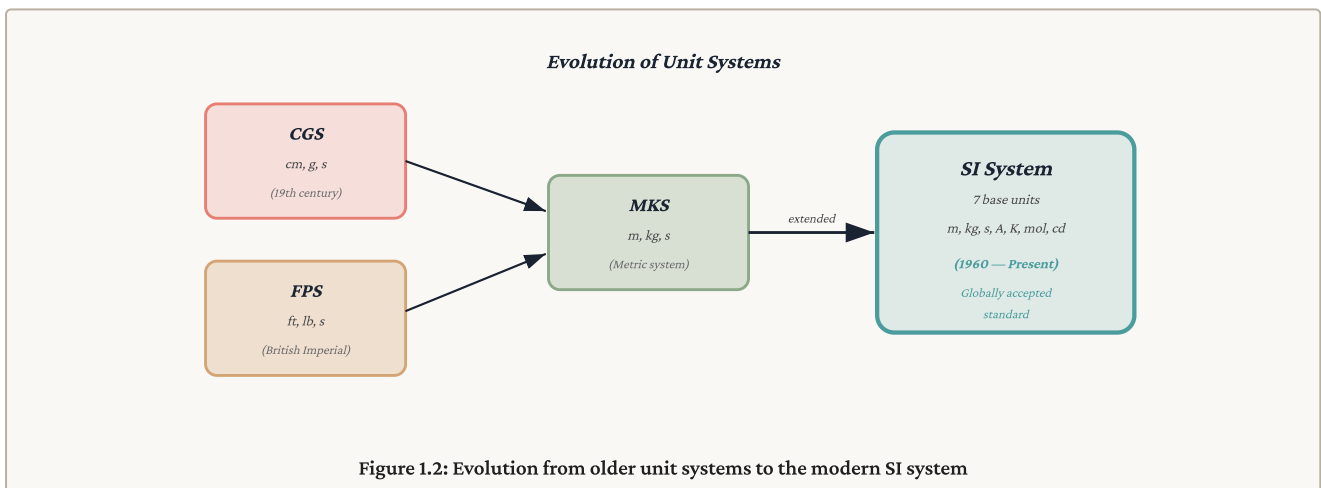
1.2.4 SI System — The Modern International System

In 1960, the General Conference on Weights and Measures (CGPM) established the **Système International d'Unités (SI)**, which has since become the globally accepted standard for scientific measurement. The SI system is an extension and refinement of the MKS system, adding units for electric current, temperature, amount of substance, and luminous intensity.

The SI system is a *coherent* system, meaning that derived units are obtained from base units without introducing numerical factors. For example, the unit of energy (joule) is exactly equal to $\text{kg m}^2 \text{s}^{-2}$ without any multiplication factor.

Advantages of the SI System

- **Universal acceptance:** Used by scientists worldwide, enabling easy communication and collaboration
- **Coherent system:** All derived units follow naturally from base units without extra conversion factors
- **Decimal-based (metric):** Conversions are powers of 10, making calculations straightforward
- **Based on physical constants:** Since 2019, all SI base units are defined using invariant physical constants (speed of light, Planck constant, etc.), ensuring long-term stability
- **Comprehensive:** Covers all branches of science — mechanics, thermodynamics, electromagnetism, optics, and chemistry
- **Regularly updated:** The CGPM revises definitions to incorporate the latest scientific understanding
- **Rational system:** Unit relationships are logically consistent across all physical quantities



1.3 SI Base Quantities and Units

1.3.1 The Seven SI Base Units

The SI system is built upon seven base (fundamental) quantities, each with a precisely defined base unit. All other physical quantities and their units are derived from these seven.

S.No.	Base Quantity	SI Unit	Symbol	Definition (Modern — 2019 Redefinition)
1	Length	metre	m	Distance light travels in vacuum in $1/299,792,458$ of a second
2	Mass	kilogram	kg	Defined by fixing the Planck constant $h = 6.62607015 \times 10^{-34} \text{ J s}$
3	Time	second	s	Duration of 9,192,631,770 periods of Cs-133 hyperfine transition
4	Electric Current	ampere	A	Defined by fixing the elementary charge $e = 1.602176634 \times 10^{-19} \text{ C}$
5	Temperature	kelvin	K	Defined by fixing the Boltzmann constant $k = 1.380649 \times 10^{-23} \text{ J/K}$
6	Amount of Substance	mole	mol	Contains exactly $6.02214076 \times 10^{23}$ elementary entities (Avogadro's number)
7	Luminous Intensity	candela	cd	Luminous efficacy of 540 THz radiation is 683 lm/W

Evolution of the Metre — From Earth to Light

- **1791:** Defined as $1/10,000,000$ of the distance from North Pole to Equator through Paris (measured by Delambre and Méchain in a 7-year survey)
- **1889:** Redefined as the distance between two marks on a platinum-iridium bar kept at the International Bureau of Weights and Measures (BIPM) in Sèvres, France
- **1960:** Redefined in terms of $1,650,763.73$ wavelengths of orange-red light from krypton-86 — the first non-artefact definition
- **1983:** Current definition adopted — the distance light travels in vacuum in $1/299,792,458$ of a second. This effectively fixes the speed of light as an exact value
- **2019:** All seven SI base units redefined in terms of fixed fundamental constants, completing the journey from physical artefacts to universal constants of nature

1.3.2 Supplementary Units

In addition to the seven base units, the SI system originally defined two **supplementary units** for measuring plane angle and solid angle. In 1995, the CGPM decided to treat these as "dimensionless derived units" rather than a separate category, but they remain widely used and important.

Definitions

Radian (rad): The radian is the plane angle subtended at the centre of a circle by an arc whose length is equal to the radius of the circle.

Steradian (sr): The steradian is the solid angle subtended at the centre of a sphere by a surface area of the sphere equal to the square of the radius.

Supplementary Quantity	Unit	Symbol	Formula
Plane Angle	radian	rad	$\theta = \text{arc length} / \text{radius} = s/r$
Solid Angle	steradian	sr	$\Omega = \text{surface area} / \text{radius}^2 = A/r^2$

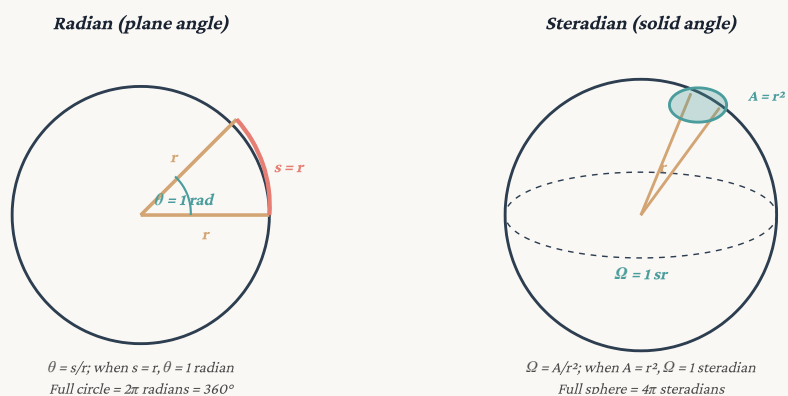


Figure 1.3: Visual representation of the radian (plane angle) and steradian (solid angle)

💡 Solved Example 1.1

Q: An arc of length 22 cm subtends an angle at the centre of a circle of radius 7 cm. Find the angle in radians and degrees.

Solution:

$$\theta = s/r = 22/7 = \pi \approx 3.14 \text{ rad}$$

Converting to degrees: $\theta = \pi \text{ rad} = 180^\circ$

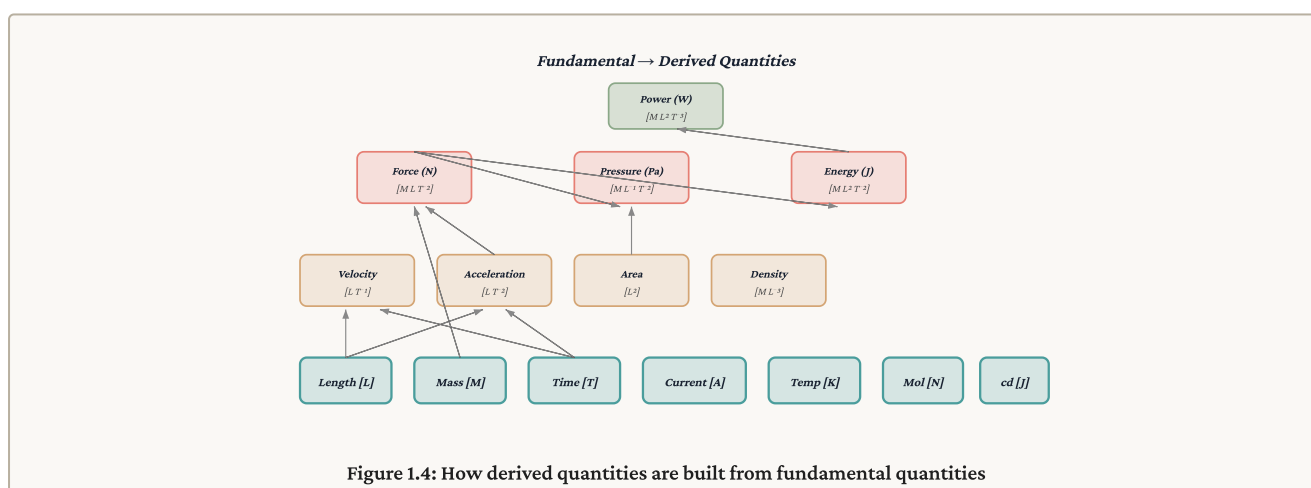
Answer: $\theta = \pi \text{ rad} = 180^\circ$ (the arc is a semicircle)

1.4 Derived Units

Derived units are obtained by combining the seven SI base units through multiplication and division. Some derived units are given special names for convenience.

1.4.1 Common Derived Units

Derived Quantity	Relation	SI Unit	Special Name	Symbol
Area	Length $\times$ Length	m^2	—	—
Volume	Length ³	m^3	—	—
Density	Mass / Volume	kg m^{-3}	—	—
Velocity	Displacement / Time	m s^{-1}	—	—
Acceleration	Velocity / Time	m s^{-2}	—	—
Force	Mass $\times$ Acceleration	kg m s^{-2}	newton	N
Pressure / Stress	Force / Area	$\text{kg m}^{-1} \text{s}^{-2}$	pascal	Pa
Energy / Work	Force $\times$ Distance	$\text{kg m}^2 \text{s}^{-2}$	joule	J
Power	Work / Time	$\text{kg m}^2 \text{s}^{-3}$	watt	W
Frequency	1 / Time	s^{-1}	hertz	Hz
Electric Charge	Current $\times$ Time	A s	coulomb	C
Potential Difference	Work / Charge	$\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$	volt	V



1.4.2 Unit Conversion

Converting between different units is a fundamental skill in physics. The key principle is: *multiplying by a conversion factor equal to 1 does not change the value of a quantity.*

Common Conversion Factors

- $1 \text{ km} = 1000 \text{ m} = 10^3 \text{ m}$
- $1 \text{ m} = 100 \text{ cm} = 10^2 \text{ cm}$
- $1 \text{ cm} = 10 \text{ mm} = 10^{-2} \text{ m}$
- $1 \text{ kg} = 1000 \text{ g} = 10^3 \text{ g}$
- $1 \text{ hour} = 3600 \text{ s}$
- $1 \text{ N} = 10^5 \text{ dyne}$
- $1 \text{ J} = 10^7 \text{ erg}$
- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$
- $1 \text{ litre} = 10^{-3} \text{ m}^3 = 1000 \text{ cm}^3$

Solved Example 1.2 — Unit Conversion

Q: Convert the speed of light, $c = 3 \times 10^8 \text{ m/s}$, to km/h .

Solution:

$$c = 3 \times 10^8 \text{ m/s}$$

We need to convert $\text{m} \rightarrow \text{km}$ and $\text{s} \rightarrow \text{h}$:

$$c = 3 \times 10^8 \times (1 \text{ km} / 1000 \text{ m}) \times (3600 \text{ s} / 1 \text{ h})$$

$$c = 3 \times 10^8 \times (1/10^3) \times 3600 \text{ km/h}$$

$$c = 3 \times 10^8 \times 3.6 \text{ km/h}$$

$$\mathbf{c = 1.08 \times 10^9 \text{ km/h}}$$

Solved Example 1.3 — CGS to SI Conversion

Q: Convert 1 dyne into newton.

Solution:

$$1 \text{ dyne} = 1 \text{ g cm s}^{-2}$$

$$= 1 \times (10^{-3} \text{ kg}) \times (10^{-2} \text{ m}) \times \text{s}^{-2}$$

$$= 10^{-5} \text{ kg m s}^{-2}$$

$$\mathbf{= 10^{-5} \text{ N}}$$

Therefore, $\mathbf{1 \text{ N} = 10^5 \text{ dyne}}$

1.4.3 SI Prefixes

Physical quantities vary enormously in magnitude. To handle such vast ranges, the SI system uses prefixes that represent powers of ten.

Prefix	Symbol	Power of 10	Numerical Value	Example
tera	T	10^{12}	1,000,000,000,000	$1 \text{ TW} = 10^{12} \text{ W}$
giga	G	10^9	1,000,000,000	$1 \text{ GHz} = 10^9 \text{ Hz}$
mega	M	10^6	1,000,000	$1 \text{ MW} = 10^6 \text{ W}$
kilo	k	10^3	1,000	$1 \text{ km} = 10^3 \text{ m}$
hecto	h	10^2	100	$1 \text{ hPa} = 100 \text{ Pa}$
deca	da	10^1	10	$1 \text{ dam} = 10 \text{ m}$
deci	d	10^{-1}	0.1	$1 \text{ dm} = 0.1 \text{ m}$
centi	c	10^{-2}	0.01	$1 \text{ cm} = 0.01 \text{ m}$
milli	m	10^{-3}	0.001	$1 \text{ mm} = 10^{-3} \text{ m}$
micro	μ	10^{-6}	0.000001	$1 \text{ }\mu\text{m} = 10^{-6} \text{ m}$
nano	n	10^{-9}	0.000000001	$1 \text{ nm} = 10^{-9} \text{ m}$
pico	p	10^{-12}	0.000000000001	$1 \text{ ps} = 10^{-12} \text{ s}$

Prefix	Symbol	Power of 10	Numerical Value	Example
femto	f	10^{-15}	0.000000000000001	1 fm = 10^{-15} m

Solved Example 1.4

Q: The mass of a box is 2.3 kg. Express this mass in (a) grams, (b) milligrams, (c) micrograms.

Solution:

(a) $2.3 \text{ kg} = 2.3 \times 10^3 \text{ g} = \mathbf{2300 \text{ g}}$

(b) $2.3 \text{ kg} = 2.3 \times 10^6 \text{ mg} = \mathbf{2.3 \times 10^6 \text{ mg}}$

(c) $2.3 \text{ kg} = 2.3 \times 10^9 \mu\text{g} = \mathbf{2.3 \times 10^9 \mu\text{g}}$

1.4.4 Order of Magnitude

Definition

Order of Magnitude: The power of 10 nearest to a quantity when expressed in scientific notation. It gives a rough estimate of the size of a quantity. If a number is expressed as $a \times 10^n$ (where $1 \leq a < 10$), the order of magnitude is 10^n if $a \leq 5$, and 10^{n+1} if $a > 5$.

For example: diameter of Earth $\approx 1.28 \times 10^7 \text{ m} \rightarrow$ order of magnitude = 10^7 m . Mass of electron $\approx 9.1 \times 10^{-31} \text{ kg} \rightarrow$ order of magnitude = 10^{-30} kg (since $9.1 > 5$). Orders of magnitude are useful for quick comparisons and estimations.

1.4.5 Range of Physical Quantities in Nature

Physical quantities in nature span an astonishing range. The following tables show the range of lengths, masses, and times found in the universe:

Object	Length (m)
Size of a proton	$\sim 10^{-15}$
Size of an atom	$\sim 10^{-10}$
Wavelength of visible light	$\sim 10^{-7}$ (400-700 nm)
Thickness of paper	$\sim 10^{-4}$
Height of a person	$\sim 10^0$ (1-2 m)
Radius of Earth	$\sim 10^7$
Earth-Sun distance	$\sim 10^{11}$
Distance to nearest star	$\sim 10^{16}$
Size of observable universe	$\sim 10^{26}$

Object	Mass (kg)
Electron	$\sim 10^{-30}$
Proton/Neutron	$\sim 10^{-27}$
Red blood cell	$\sim 10^{-14}$
Mosquito	$\sim 10^{-5}$
Human	$\sim 10^2$ (50-100 kg)
Earth	$\sim 10^{25}$
Sun	$\sim 10^{30}$
Milky Way galaxy	$\sim 10^{41}$

Event	Time (s)
Time for light to cross nucleus	$\sim 10^{-23}$

Event	Time (s)
Period of visible light wave	$\sim 10^{-15}$
Period of audible sound	$\sim 10^{-3}$
Human heartbeat	$\sim 10^0$ (1 s)
One day	$\sim 10^5$
One year	$\sim 10^7$
Human lifespan	$\sim 10^9$
Age of universe	$\sim 10^{17}$

1.5 Errors in Measurement

No measurement is perfect. Every measuring instrument has limitations, and every measurement involves some degree of uncertainty. The difference between the measured value and the true value of a quantity is called an **error**. Understanding errors is crucial for interpreting experimental results.

Who Invented Error Analysis?

Carl Friedrich Gauss (1777–1855), while working on astronomical observations, developed the mathematical theory of errors. He showed that random measurement errors follow a bell-shaped curve — the famous **Gaussian (normal) distribution** — and that the arithmetic mean is the best estimate of the true value. His method of *least squares* (1795) remains the foundation of data analysis today.

Pierre-Simon Laplace (1749–1827) independently contributed to the probability theory underlying error analysis. Together, Gauss and Laplace transformed measurement from a craft into a rigorous science, giving us the tools (mean, standard deviation, probability distributions) that every scientist uses today.

1.5.1 Systematic Errors

Definition

Systematic Error: Errors that consistently occur in the same direction (always too high or always too low) are called systematic errors. They affect the *accuracy* of measurement and tend to shift all measurements by a fixed amount or percentage.

Types of Systematic Errors

- **Instrumental errors:** Arise from imperfect design, calibration, or zero error of the instrument. Example: a metre scale that is slightly shorter than 1 m, a voltmeter with a zero error
- **Imperfection in experimental technique:** Arise from the method or setup used. Example: measuring body temperature by placing thermometer under the arm instead of in the mouth
- **Personal errors (observational):** Arise from the observer's habits. Example: always reading a scale from an angle (parallax error), reaction time delay in starting/stopping a stopwatch
- **Environmental errors:** Arise from external conditions. Example: temperature changes causing expansion of a scale, humidity affecting electrical measurements

How to minimize: Systematic errors can be reduced by (i) improving experimental techniques, (ii) using more precise instruments, (iii) calibrating instruments before use, (iv) removing personal bias, and (v) repeating measurements using different methods.

1.5.2 Random Errors

Definition

Random Error: Errors that fluctuate unpredictably from one measurement to another, sometimes giving a value that is too high and sometimes too low. They affect the *precision* of measurements and arise from unknown and unpredictable variations.

Causes of Random Errors

- Small, unpredictable fluctuations in experimental conditions
- Limitations in reading analog instruments to the finest division
- Observer's varying judgment when estimating between scale divisions
- Vibrations, electrical noise, and other uncontrollable disturbances

How to minimize: Random errors can be reduced by taking a large number of measurements and calculating their arithmetic mean. The mean value is closer to the true value than any individual measurement. By statistics, the error in the mean decreases as $1/\sqrt{n}$, where n is the number of observations.

1.5.3 Least Count Error

Definition

Least Count: The least count of an instrument is the smallest measurement that can be made using that instrument. It equals the value of one smallest division on the scale.

Least Count Error: The error associated with the resolution of the measuring instrument. It is the smallest value that can be measured and sets the lower limit on the precision of a measurement.

LEAST COUNT FORMULAS

Vernier Caliper: $L.C. = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ MSD} / n$ (where n = number of vernier divisions)

Screw Gauge: $L.C. = \text{Pitch} / \text{Number of circular scale divisions}$

Measuring Instrument	Typical Least Count	Used For
Metre Scale	1 mm (0.1 cm)	Measuring lengths of everyday objects
Vernier Caliper	0.1 mm (0.01 cm)	Precise measurement of diameters, depths
Micrometer Screw Gauge	0.01 mm (10 μm)	Very precise measurements of thin objects
Digital Balance	0.01 g or 0.001 g	Measuring mass accurately
Stopwatch (analog)	0.1 s	Measuring time intervals
Digital Stopwatch	0.01 s	Precise time measurement

Solved Example 1.5

Q: A vernier caliper has 1 mm divisions on the main scale and 10 divisions on the vernier scale. Calculate its least count. If the main scale reading is 2.3 cm and vernier coincidence is 7, what is the total reading?

Solution:

$$1 \text{ MSD} = 1 \text{ mm} = 0.1 \text{ cm}$$

$$L.C. = 1 \text{ MSD} / 10 = 0.1/10 = \mathbf{0.01 \text{ cm}}$$

$$\text{Total reading} = \text{MSR} + (\text{VSD coincidence} \times L.C.)$$

$$= 2.3 + (7 \times 0.01)$$

$$= 2.3 + 0.07$$

$$= \mathbf{2.37 \text{ cm}}$$

1.5.4 Absolute, Relative, and Percentage Error

To quantify errors precisely, we use three important measures: absolute error, relative error (also called fractional error), and percentage error.

Definitions

True value (a_{mean}): In the absence of a known true value, the arithmetic mean of multiple measurements is taken as the best estimate of the true value:

$$a_{\text{mean}} = (a_1 + a_2 + a_3 + \dots + a_n) / n$$

Absolute Error (Δa_i): The magnitude of the difference between the individual measurement and the true (mean) value:

$$\Delta a_i = |a_i - a_{\text{mean}}|$$

Mean Absolute Error (Δa_{mean}): The arithmetic mean of all absolute errors:

$$\Delta a_{\text{mean}} = (|\Delta a_1| + |\Delta a_2| + \dots + |\Delta a_n|) / n$$

Relative Error (or Fractional Error): The ratio of mean absolute error to the mean value:

$$\text{Relative Error} = \Delta a_{\text{mean}} / a_{\text{mean}}$$

Percentage Error: The relative error expressed as a percentage:

$$\text{Percentage Error} = (\Delta a_{\text{mean}} / a_{\text{mean}}) \times 100\%$$

ERROR FORMULAS SUMMARY

$$\text{Absolute Error: } \Delta a_i = |a_i - a_{\text{mean}}|$$

$$\text{Mean Absolute Error: } \Delta a_{\text{mean}} = (\sum |\Delta a_i|) / n$$

$$\text{Relative Error} = \Delta a_{\text{mean}} / a_{\text{mean}}$$

$$\text{Percentage Error} = (\Delta a_{\text{mean}} / a_{\text{mean}}) \times 100\%$$

$$\text{Final result is expressed as: } a = a_{\text{mean}} \pm \Delta a_{\text{mean}}$$

Solved Example 1.6 — Complete Error Analysis

Q: The diameter of a wire was measured five times and the readings were: 1.52 mm, 1.50 mm, 1.54 mm, 1.53 mm, 1.51 mm. Find (a) mean value, (b) absolute errors, (c) mean absolute error, (d) relative error, and (e) percentage error.

Solution:

(a) Mean value:

$$a_{\text{mean}} = (1.52 + 1.50 + 1.54 + 1.53 + 1.51) / 5 = 7.60 / 5 = \mathbf{1.52 \text{ mm}}$$

(b) Absolute errors:

$$\Delta a_1 = |1.52 - 1.52| = 0.00 \text{ mm}$$

$$\Delta a_2 = |1.50 - 1.52| = 0.02 \text{ mm}$$

$$\Delta a_3 = |1.54 - 1.52| = 0.02 \text{ mm}$$

$$\Delta a_4 = |1.53 - 1.52| = 0.01 \text{ mm}$$

$$\Delta a_5 = |1.51 - 1.52| = 0.01 \text{ mm}$$

(c) Mean absolute error:

$$\Delta a_{\text{mean}} = (0.00 + 0.02 + 0.02 + 0.01 + 0.01) / 5 = 0.06/5 = \mathbf{0.012 \text{ mm} \approx 0.01 \text{ mm}}$$

(d) Relative error:

$$= \Delta a_{\text{mean}} / a_{\text{mean}} = 0.01 / 1.52 = \mathbf{0.0066}$$

(e) Percentage error:

$$= 0.0066 \times 100\% = \mathbf{0.66\%}$$

Final result: Diameter = $1.52 \pm 0.01 \text{ mm}$

1.5.5 Propagation of Errors in Calculations

When we use measured quantities in calculations, the errors propagate. The rules for error propagation depend on the mathematical operation.

Rules for Combination of Errors

- **Addition or Subtraction ($Z = A \pm B$):** The maximum absolute error in Z is the sum of absolute errors: $\Delta Z = \Delta A + \Delta B$
- **Multiplication or Division ($Z = A \times B$ or $Z = A/B$):** The maximum relative error in Z is the sum of relative errors: $\Delta Z/Z = \Delta A/A + \Delta B/B$
- **Power ($Z = A^n$):** The relative error in Z is n times the relative error in A: $\Delta Z/Z = n \times (\Delta A/A)$
- **General formula ($Z = A^p B^q / C^r$):** $\Delta Z/Z = p(\Delta A/A) + q(\Delta B/B) + r(\Delta C/C)$

Solved Example 1.7 — Error Propagation

Q: The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{L/g}$. Calculate the percentage error in T if L is measured with 2% error and g with 1% error.

Solution:

$$T = 2\pi(L/g)^{1/2} = 2\pi L^{1/2} g^{-1/2}$$

Using the power rule:

$$\Delta T/T = (1/2)(\Delta L/L) + (1/2)(\Delta g/g)$$

$$\text{Percentage error in } T = (1/2)(2\%) + (1/2)(1\%)$$

$$= 1\% + 0.5\%$$

$$= \mathbf{1.5\%}$$

💡 Solved Example 1.8 — Error in Density

Q: The mass of a sphere is measured as (2.00 ± 0.01) kg and its radius as (0.10 ± 0.01) m. Find the percentage error in the density.

Solution:

$$\text{Density } \rho = \text{Mass} / \text{Volume} = M / (4/3 \pi r^3) \rightarrow \rho \propto M r^{-3}$$

$$\Delta\rho/\rho = \Delta M/M + 3(\Delta r/r)$$

$$= (0.01/2.00) + 3 \times (0.01/0.10)$$

$$= 0.005 + 3 \times 0.1$$

$$= 0.005 + 0.3 = 0.305$$

Percentage error = 30.5%

Note: The error is dominated by the radius measurement because it appears cubed. This shows why the least accurate measurement with the highest power contributes most to the error.

1.6 Accuracy vs Precision

📖 Key Definitions

Accuracy: How close a measured value is to the *true value* of the quantity being measured. An accurate measurement has minimal *systematic error*. Accuracy tells us about the *correctness* of a measurement.

Precision: How close *repeated measurements* of the same quantity are to *each other*. A precise measurement has minimal *random error* and high *reproducibility*. Precision tells us about the *resolution* of a measurement.

It is critical to understand that accuracy and precision are **independent** concepts. A measurement can be precise but not accurate, accurate but not precise, both, or neither.

Accuracy vs Precision — Target Analogy

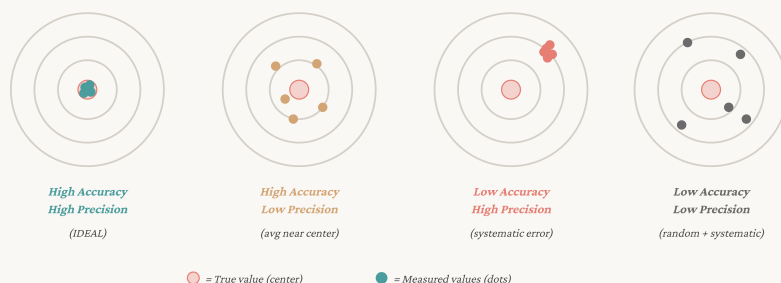


Figure 1.5: Target analogy showing the four combinations of accuracy and precision

💡 Illustrative Example

Suppose the actual length of a pencil is **15.0 cm**. Four students measure it:

Student	Readings (cm)	Mean	Accuracy	Precision
A	15.1, 15.0, 14.9	15.0	High ✓	High ✓
B	17.2, 17.3, 17.1	17.2	Low ✗	High ✓
C	14.5, 15.8, 14.2	14.8	Low ✗	Low ✗
D	16.2, 13.8, 15.1	15.0	High ✓	Low ✗

Student B likely has a systematic error (e.g., zero error in the scale). **Student D** is accidentally accurate on average, but unreliable for any single measurement.

Feature	Accuracy	Precision
Relates to	True value	Repeated measurements (each other)

Feature	Accuracy	Precision
Affected by	Systematic errors	Random errors
Improved by	Better technique, calibration	Averaging multiple readings
Tells us	How correct the result is	How reproducible the result is
Can exist without the other?	Yes	Yes

1.7 Significant Figures

When we write a measured value, the number of digits we use conveys information about the precision of the measurement. **Significant figures** (also called significant digits) are the digits in a number that carry meaningful information about its precision. They include all *certain* digits plus the first *uncertain* (estimated) digit.

1.7.1 Rules for Counting Significant Figures

Rules for Identifying Significant Figures

- **Rule 1:** All non-zero digits are always significant.
Example: 247 → 3 sig figs; 1.52 → 3 sig figs
- **Rule 2:** Zeros between non-zero digits (captive zeros) are significant.
Example: 1005 → 4 sig figs; 2.03 → 3 sig figs
- **Rule 3:** Leading zeros (zeros before the first non-zero digit) are *never* significant. They merely indicate the position of the decimal point.
Example: 0.0025 → 2 sig figs; 0.00340 → 3 sig figs
- **Rule 4:** Trailing zeros in a number with a decimal point *are* significant.
Example: 2.500 → 4 sig figs; 0.0200 → 3 sig figs; 100.0 → 4 sig figs
- **Rule 5:** Trailing zeros in a whole number without a decimal point are ambiguous. Use scientific notation to clarify.
Example: 1500 → ambiguous (2, 3, or 4 sig figs); 1.500×10^3 → 4 sig figs
- **Rule 6:** Exact numbers (counted quantities, defined constants) have *infinite* significant figures.
Example: 12 inches = 1 foot (exact); $\pi \approx 3.14159...$ (mathematical constant)

Number	Significant Figures	Explanation
234	3	All non-zero digits
2.34	3	All non-zero digits
0.0025	2	Leading zeros don't count (only 2 and 5)
2.005	4	Zeros between non-zero digits count
3.400	4	Trailing zeros after decimal count
0.00340	3	Leading zeros don't count; trailing zero counts
2.40×10^3	3	Scientific notation clarifies sig figs
5000	Ambiguous (1–4)	Use scientific notation to clarify
5.000×10^3	4	All digits in coefficient are significant

1.7.2 Rounding Off

Rounding Rules

- **Rule A:** If the digit to be dropped is **less than 5**, leave the preceding digit unchanged.
Example: 2.34 rounded to 2 sig figs → **2.3**
- **Rule B:** If the digit to be dropped is **greater than 5**, increase the preceding digit by 1.
Example: 2.36 rounded to 2 sig figs → **2.4**
- **Rule C:** If the digit to be dropped is **exactly 5**, round to the nearest *even* number (banker's rounding).
Example: 2.35 → **2.4** (round up to even); 2.45 → **2.4** (already even); 3.55 → **3.6**

1.7.3 Significant Figures in Arithmetic Operations

Addition and Subtraction

The result should have the same number of **decimal places** as the measurement with the *fewest* decimal places.

Solved Example 1.9

Q: Add 25.2 cm + 3.87 cm + 0.456 cm with correct significant figures.

Solution:

$$25.2 + 3.87 + 0.456 = 29.526 \text{ cm}$$

25.2 has only 1 decimal place (least), so the answer should have 1 decimal place.

Answer: 29.5 cm

Multiplication and Division

The result should have the same number of **significant figures** as the factor with the *fewest* significant figures.

Solved Example 1.10

Q: The length, breadth, and thickness of a rectangular block are 25.2 cm, 10.1 cm, and 5.2 cm. Find its volume with correct significant figures.

Solution:

$$\text{Volume} = l \times b \times t = 25.2 \times 10.1 \times 5.2 = 1322.496 \text{ cm}^3$$

Significant figures: 25.2 → 3; 10.1 → 3; 5.2 → 2 (least)

Result should have 2 significant figures.

Volume = 1300 cm³ = 1.3 × 10³ cm³

Important Note

Never round intermediate results during a multi-step calculation. Only round the **final answer**. Rounding intermediate values leads to accumulation of rounding errors.

1.7.4 Scientific Notation (Standard Form)

Definition

Scientific Notation: A way of writing numbers in the form $a \times 10^n$, where $1 \leq a < 10$ and n is an integer. This notation clearly shows the number of significant figures and is convenient for very large or very small numbers.

Advantages of Scientific Notation

- Makes it easy to express very large or very small numbers compactly
- Removes ambiguity about the number of significant figures
- Simplifies arithmetic operations (multiply/divide coefficients, add/subtract exponents)
- Makes comparison of magnitudes straightforward

1.8.3 Dimensional Formulas of Important Physical Quantities

Physical Quantity	Formula	Dimensional Formula	SI Unit
Length	Fundamental	$[M^0 L^1 T^0] = [L]$	m
Mass	Fundamental	$[M^1 L^0 T^0] = [M]$	kg
Time	Fundamental	$[M^0 L^0 T^1] = [T]$	s
Area	$L \times L$	$[L^2]$	m^2
Volume	$L \times L \times L$	$[L^3]$	m^3
Density	Mass / Volume	$[M L^{-3}]$	$kg\ m^{-3}$
Velocity	Displacement / Time	$[L T^{-1}]$	$m\ s^{-1}$
Acceleration	Velocity / Time	$[L T^{-2}]$	$m\ s^{-2}$
Momentum	Mass $\times$ Velocity	$[M L T^{-1}]$	$kg\ m\ s^{-1}$
Force	Mass $\times$ Acceleration	$[M L T^{-2}]$	N (newton)
Impulse	Force $\times$ Time	$[M L T^{-1}]$	N s
Work / Energy	Force $\times$ Distance	$[M L^2 T^{-2}]$	J (joule)
Power	Work / Time	$[M L^2 T^{-3}]$	W (watt)
Pressure / Stress	Force / Area	$[M L^{-1} T^{-2}]$	Pa (pascal)
Torque	Force $\times$ Distance	$[M L^2 T^{-2}]$	N m
Surface Tension	Force / Length	$[M T^{-2}]$	$N\ m^{-1}$
Coefficient of Viscosity	$F / (A \times dv/dx)$	$[M L^{-1} T^{-1}]$	Pa s
Gravitational Constant (G)	$Fr^2 / (m_1 m_2)$	$[M^{-1} L^3 T^{-2}]$	$N\ m^2\ kg^{-2}$
Frequency	1 / Time	$[T^{-1}]$	Hz (hertz)
Angle (plane)	Arc / Radius	$[M^0 L^0 T^0]$ (dimensionless)	rad
Strain	$\Delta L / L$	$[M^0 L^0 T^0]$ (dimensionless)	—

Solved Example 1.12 — Deriving Dimensional Formula

Q: Derive the dimensional formula of (a) Gravitational constant G, (b) Coefficient of viscosity η , and (c) Surface tension σ .

(a) Gravitational constant G:

From $F = Gm_1m_2/r^2$, we get $G = Fr^2/(m_1m_2)$

$$[G] = [M L T^{-2}][L^2] / [M][M] = [M L^3 T^{-2}] / [M^2]$$

$$[G] = [M^{-1} L^3 T^{-2}]$$

(b) Coefficient of viscosity η :

From $F = \eta A(dv/dx)$, we get $\eta = F/(A \times dv/dx)$

$$[\eta] = [M L T^{-2}] / ([L^2] \times [L T^{-1}]/[L]) = [M L T^{-2}] / ([L^2] \times [T^{-1}])$$

$$[\eta] = [M L T^{-2}] / [L^2 T^{-1}]$$

$$[\eta] = [M L^{-1} T^{-1}]$$

(c) Surface tension σ :

Surface tension = Force / Length

$$[\sigma] = [M L T^{-2}] / [L]$$

$$[\sigma] = [M T^{-2}]$$

Important: Dimensionless Quantities

Some physical quantities have no dimensions (dimensional formula $[M^0 L^0 T^0]$). These include:

- Angle (radian) — arc/radius = L/L
- Strain — $\Delta L/L$
- Refractive index — c/v (velocity/velocity)
- Relative density — ρ/ρ_{water}
- Pure numbers (π , e , etc.)
- Trigonometric ratios ($\sin \theta$, $\cos \theta$, etc.)

1.9 Dimensional Analysis

Dimensional analysis is a powerful method that uses dimensions to check equations, derive relations, and convert units. It is based on the fact that physical equations must be dimensionally consistent.



Joseph Fourier — Pioneer of Dimensional Analysis

1768–1830, France

In his 1822 masterwork *Théorie analytique de la chaleur* (Analytical Theory of Heat), Fourier was the first to systematically use the principle of dimensional consistency. He insisted that every term in a physical equation must have the same dimensions — the principle of homogeneity — and used it to check equations and derive new results. This was the birth of dimensional analysis as a formal technique.



Lord Rayleigh (John William Strutt) — Master of Dimensional Reasoning

1842–1919, England

Lord Rayleigh (Nobel Prize 1904) elevated dimensional analysis into a powerful problem-solving tool. He used it extensively to derive relationships in acoustics, optics, and fluid mechanics — often arriving at correct formulas (up to dimensionless constants) without solving any differential equations. His approach demonstrated that dimensional analysis is not just a checking tool but a genuine method of discovery.

1.9.1 Principle of Homogeneity of Dimensions

Principle

Principle of Homogeneity: In any correct physical equation, the dimensions of all terms on both sides of the equation must be the same. If $[A] = [B] + [C]$, then $[A] = [B] = [C]$. You can only add, subtract, or equate quantities that have the same dimensions.

1.9.2 Application 1: Checking the Correctness of Equations

If an equation is dimensionally incorrect, it is definitely wrong. However, dimensional correctness does *not* guarantee that the equation is completely correct — numerical constants and dimensionless factors cannot be checked.



Solved Example 1.13 — Checking Dimensional Consistency

Q: Check whether the equation $v^2 = u^2 + 2as$ is dimensionally correct.

Solution:

$$\text{LHS: } [v^2] = [L T^{-1}]^2 = [L^2 T^{-2}]$$

$$\text{RHS, first term: } [u^2] = [L T^{-1}]^2 = [L^2 T^{-2}] \quad \checkmark$$

$$\text{RHS, second term: } [2as] = [L T^{-2}][L] = [L^2 T^{-2}] \quad \checkmark$$

All three terms have the same dimensions $[L^2 T^{-2}]$.

The equation is dimensionally consistent. ✓

💡 Solved Example 1.14

Q: Verify the dimensional consistency of $s = ut + \frac{1}{2}at^2$.

Solution:

LHS: $[s] = [L]$

RHS term 1: $[ut] = [L T^{-1}] \times [T] = [L] \checkmark$

RHS term 2: $[\frac{1}{2}at^2] = [L T^{-2}] \times [T^2] = [L] \checkmark$

(Note: $\frac{1}{2}$ is dimensionless and does not affect dimensions)

All terms have dimensions $[L]$. **The equation is dimensionally consistent. $\checkmark$**

1.9.3 Application 2: Converting Units from One System to Another

Dimensional analysis provides a systematic method for converting units between different systems (e.g., CGS to SI).

UNIT CONVERSION USING DIMENSIONAL ANALYSIS

If a quantity has dimensions $[M^a L^b T^c]$, then:

$$n_2 = n_1 \times (M_1/M_2)^a \times (L_1/L_2)^b \times (T_1/T_2)^c$$

where n_1, n_2 are numerical values in systems 1 and 2;

M_1, L_1, T_1 and M_2, L_2, T_2 are the fundamental units in systems 1 and 2.

💡 Solved Example 1.15 — Unit Conversion by Dimensional Analysis

Q: Convert 1 joule into erg using dimensional analysis.

Solution:

Energy has dimensions $[M L^2 T^{-2}]$, so $a = 1, b = 2, c = -2$

System 1 (SI): $M_1 = 1 \text{ kg}, L_1 = 1 \text{ m}, T_1 = 1 \text{ s}, n_1 = 1$

System 2 (CGS): $M_2 = 1 \text{ g}, L_2 = 1 \text{ cm}, T_2 = 1 \text{ s}$

$$n_2 = n_1 \times (M_1/M_2)^1 \times (L_1/L_2)^2 \times (T_1/T_2)^{-2}$$

$$= 1 \times (1 \text{ kg} / 1 \text{ g})^1 \times (1 \text{ m} / 1 \text{ cm})^2 \times (1 \text{ s} / 1 \text{ s})^{-2}$$

$$= 1 \times (1000)^1 \times (100)^2 \times (1)^{-2}$$

$$= 1 \times 1000 \times 10000 \times 1$$

$$= 10^7$$

Therefore, **1 joule = 10^7 erg**

1.9.4 Application 3: Deriving Relations Between Physical Quantities

When we know which physical quantities a given quantity depends on, we can use dimensional analysis to derive the mathematical relationship between them.

Steps for Deriving Relations by Dimensional Analysis



Example: Deriving Time Period of Simple Pendulum

Step 1: T (time period) is the dependent quantity

Step 2: T may depend on: length (l), mass (m), acceleration due to gravity (g)

$$\text{Step 3: } T = k l^a m^b g^c$$

$$\text{Step 4: } [T] = [L]^a [M]^b [LT^{-2}]^c \rightarrow [M^0 L^1 T^1] = [M^b L^{a+c} T^{-2c}]$$

$$\text{Step 5: } b = 0, a + c = 0, -2c = 1 \rightarrow c = -\frac{1}{2}, a = \frac{1}{2}$$

$$\text{Result: } T = k \sqrt{l/g} \text{ (} k = 2\pi \text{ found experimentally)}$$

Figure 1.6: Step-by-step process for deriving relations using dimensional analysis

Solved Example 1.16 — Deriving Formula by Dimensional Analysis

Q: Derive the formula for the time period of a simple pendulum using dimensional analysis.

Solution:

The time period T may depend on: length l , mass m , and acceleration due to gravity g .

Let $T = k l^a m^b g^c$ (where k is a dimensionless constant)

Writing dimensions on both sides:

$$[M^0 L^0 T^1] = [L]^a [M]^b [L T^{-2}]^c$$

$$[M^0 L^0 T^1] = [M^b L^{a+c} T^{-2c}]$$

Comparing powers of M : $b = 0$

Comparing powers of T : $1 = -2c \rightarrow c = -1/2$

Comparing powers of L : $0 = a + c \rightarrow a = -c = 1/2$

Therefore: $T = k l^{1/2} m^0 g^{-1/2} = k \sqrt{l/g}$

Experimentally, the constant $k = 2\pi$, so $T = 2\pi\sqrt{l/g}$.

Note: The value of $k = 2\pi$ cannot be determined by dimensional analysis alone.

Solved Example 1.17

Q: The velocity v of a particle depends upon time t according to the equation $v = a + bt + c/t$. Find the dimensions of a , b , and c .

Solution:

By the principle of homogeneity, each term must have the same dimensions as v .

$$[v] = [L T^{-1}]$$

For a : $[a] = [v] = [L T^{-1}]$ (same as velocity)

For bt : $[bt] = [v] \rightarrow [b][T] = [L T^{-1}] \rightarrow [b] = [L T^{-2}]$ (same as acceleration)

For c/t : $[c/t] = [v] \rightarrow [c]/[T] = [L T^{-1}] \rightarrow [c] = [L]$ (same as length)

Solved Example 1.18

Q: In the equation $P = (a - t^2)/(bx)$, where P is pressure, t is time, and x is distance, find the dimensions of a/b .

Solution:

By homogeneity, the numerator $(a - t^2)$ must have same dimensions, so:

$$[a] = [t^2] = [T^2]$$

$$\text{Also, } [P] = [(a - t^2)] / [bx]$$

$$[M L^{-1} T^{-2}] = [T^2] / [b][L]$$

$$[b] = [T^2] / ([M L^{-1} T^{-2}][L]) = [T^2] / [M T^{-2}] = [M^{-1} T^4]$$

$$\text{Therefore: } [a/b] = [T^2] / [M^{-1} T^4] = [M T^{-2}]$$

1.9.5 Limitations of Dimensional Analysis

While powerful, dimensional analysis has important limitations that must be understood:

Limitations of Dimensional Analysis

- **Cannot determine dimensionless constants:** Constants like 2 , π , $1/2$, $1/3$, etc. cannot be found. For example, we get $T = k\sqrt{l/g}$, but $k = 2\pi$ is unknown
- **Cannot handle transcendental functions:** Equations involving trigonometric ($\sin$, $\cos$), exponential (e^x), or logarithmic ($\log x$) functions cannot be derived or checked for the argument
- **Cannot distinguish same-dimension quantities:** Work $[M L^2 T^{-2}]$ and torque $[M L^2 T^{-2}]$ have the same dimensions but are different physical quantities
- **Cannot derive relations with more than 3 unknowns** (for mechanical quantities, since we only have 3 equations from M , L , T)
- **Cannot determine the form of complex relations:** If a quantity depends on the sum or difference of two quantities, dimensional analysis cannot determine this
- **Gives no information about whether a quantity is scalar or vector**

Chapter Summary — Must Remember

- A physical quantity = Numerical value $\times$ Unit. Both are essential
- The SI system has 7 base units: m, kg, s, A, K, mol, cd
- Supplementary units: radian (plane angle) and steradian (solid angle)
- CGS uses cm-g-s; FPS uses ft-lb-s; MKS uses m-kg-s; SI extends MKS to 7 quantities
- Systematic errors affect accuracy; random errors affect precision
- Absolute error = |measured – true|; Relative error = $\Delta a/a$; % error = $(\Delta a/a) \times 100$
- Error in power: if $Z = A^n$, then $\Delta Z/Z = n(\Delta A/A)$
- Accuracy = closeness to true value; Precision = reproducibility
- Leading zeros are never significant; trailing zeros after decimal are significant
- Scientific notation: $a \times 10^n$ ($1 \leq a < 10$) removes ambiguity about sig figs
- Dimensions tell us the nature of a physical quantity in terms of M, L, T
- Dimensional formula vs. dimensional equation: formula is the expression; equation equates the quantity to it
- Dimensional analysis can: check equations, convert units, derive relations
- Dimensional analysis cannot: find dimensionless constants, handle trig/exponential functions, distinguish same-dimension quantities

Common Mistakes to Avoid

- Confusing precision with accuracy — they are independent concepts
- Counting leading zeros as significant figures
- Forgetting that angle, strain, and refractive index are dimensionless
- Using dimensional analysis to find numerical constants like π or $1/2$
- Rounding intermediate results in multi-step calculations
- Writing answers without proper units
- Confusing relative error with absolute error
- Forgetting to add relative errors (not absolute) in multiplication/division
- Forgetting the power rule: error in A^n is n times the relative error in A

Practice Questions (Unsolved)

Conceptual Questions

1. Why is measurement the foundation of physics?
2. Difference between fundamental and derived quantities? Give 3 examples each.
3. Why was SI adopted internationally? Advantages over CGS and FPS?
4. Can a dimensionally correct equation be wrong? Explain with example.
5. Distinguish between accuracy and precision with diagrams.
6. What is the difference between systematic and random errors?
7. Why can dimensional analysis not determine dimensionless constants?

Short Answer Questions (2-3 Marks)

1. State rules for significant figures. Apply to: 0.00470, 3.400, 2.05×10^3 .
2. Derive dimensional formula for pressure, energy, and gravitational constant G .
3. Using dimensional analysis, derive the time period of a simple pendulum.
4. Explain radian and steradian with diagrams.
5. Check dimensional correctness of $v = u + at^2$.

Numerical Problems

1. Convert 1 joule into erg using dimensional analysis.
2. Add 25.2 cm + 3.87 cm + 0.456 cm with correct significant figures.
3. Speed of light = 3×10^8 m/s. Convert to km/h.
4. Find % error in $T = 2\pi\sqrt{L/g}$ if L has 2% error and g has 1% error.
5. In $P = (a - t^2)/bx$ (P =pressure, t =time, x =distance), find dimensions of a and b .

6. A wire diameter measured 5 times: 1.52, 1.50, 1.54, 1.53, 1.51 mm. Find mean, absolute error, relative error, % error.
7. Convert 1 newton into dyne using dimensional analysis.

Assertion-Reason Questions

(a) Both true, R explains A (b) Both true, R not explanation (c) A true, R false (d) A false, R true

1. **A:** Work and torque have same dimensions $[ML^2T^{-2}]$. **R:** Dimensional analysis can always distinguish different quantities.
2. **A:** 0.0050 has 2 significant figures. **R:** Trailing zeros after decimal are significant; leading zeros are not.
3. **A:** A dimensionally correct equation need not be physically correct. **R:** Dimensional analysis cannot find dimensionless constants.
4. **A:** Relative errors add in multiplication. **R:** For $Z = AB$, $\Delta Z/Z = \Delta A/A + \Delta B/B$.